

Dynamic Routing and Wavelength Assignment in WDM Networks with Ant-Based Agents

Son-Hong Ngo¹, Xiaohong Jiang¹, Susumu Horiguchi¹, and Minyi Guo²

¹ Graduate School of Information Science, Japan Advanced Institute of Science and Technology, Japan
{sonhong, jiang, hori}@jaist.ac.jp

² School of Computer Science and Engineering, The University of Aizu, Japan
minyi@u-aizu.ac.jp

Abstract. In this paper, we propose an ant-based algorithm for dynamic routing and wavelength assignment (RWA) in WDM optical networks under the wavelength continuity constraint. By adopting a new routing table structure and keeping a number of ants in the network to cooperatively explore the network states and continuously update the routing tables, our new ant algorithm can efficiently support the ants' foraging tasks of route selection and wavelength assignment in WDM networks, and allow a connection to be setup promptly on arrival with a small setup time. Extensive simulation results based on the ns-2 network simulator indicate that the proposed algorithm can adapt well to traffic variations and achieves a lower blocking probability than the fixed routing algorithm.

1 Introduction

All optical networks that adopt wavelength-division-multiplexing (WDM) technology have a huge bandwidth capacity, and they show promise as the backbone of the next generation Internet. In all optical networks, data are routed in optical channels called lightpaths. The Routing and Wavelength Assignment (RWA) problem is how to determine both a route and wavelengths for a connection request. Without wavelength conversion capability, a lightpath must use the same wavelength on all the links along its route, which is referred to as the wavelength continuity constraint.

The RWA problem is usually classified as the static RWA problem and the dynamic RWA problem. In the static RWA problem, the connection requests are given in advance, and the problem becomes how to establish lightpaths for all these requests so that the total number of wavelengths is minimized. Static RWA has been proved to be an NP-complete problem [1]. In the dynamic RWA problem, the traffic is dynamic with connection requests arriving randomly, making it more difficult. Heuristic algorithms are usually employed to resolve this problem. Generally, a dynamic RWA algorithm aims to minimize the total blocking probability in the entire network.

In our work, we focus on the dynamic RWA problem with wavelength continuity constraint. In the literature, the dynamic RWA problem is usually divided into two sub-problems that can be solved separately: routing and wavelength assignment. Routing schemes can be classified into fixed routing, fixed-alternate routing and

adaptive routing. In the fixed routing scheme, one route is dedicated for a source-destination pair. Whenever a request occurs between this source-destination pair, this route is attempted for wavelength assignment. The fixed routing method is simple but usually causes a high blocking probability. The fixed-alternate routing method has better performance with multiple paths dedicated for a node pair. In the adaptive routing scheme, the route is computed at the time the connection request arrives, based on the current network state, thus it yields the best performance. However, adaptive routing requires high computational complexity. A more detailed survey of routing and wavelength assignment can be found in [2].

The adaptive RWA solutions in the literature always need special support from control protocol to obtain the global state of the network. Moreover, heuristic algorithms that perform route and wavelength searching tasks after a request arrives must take into account the tradeoff between complexity and performance. This also contributes to high setup delay and control overhead. A possible approach to overcome these problems is the use of ant-based mobile agents [3]. The ant-based agent routing approach inherits advantages from both mobile agents behaviors and an ant colony system. Recent results show that this approach could yield efficient performance in the control of both circuit switching [4] and packet switching networks [5].

In this paper, we investigate a new ant-based agent algorithm for the dynamic RWA problem in WDM networks under the constraint of wavelength continuity. Our study aims to reduce both blocking probability and path setup time by using a suitable amount of ants, which continuously perform path searching tasks before the connection request's arrival so that the route selection and wavelength assignment of a request are performed by simply looking up the routing tables. To achieve that goal, we develop a new routing table structure, a scheme for ant population control and a mechanism for pheromone updating, for our new algorithm.

The rest of this paper is organized as follows: In section 2, we discuss related works. Section 3 presents our new approach to the dynamic RWA problem in WDM networks under wavelength continuity constraint. Section 4 describes our preliminary simulation and analysis results. Finally, our conclusions and future works are discussed in Section 5

2 Related Work

Recent research results show that the routing in communication networks can be resolved efficiently by means of Ant Colony Optimization (ACO) [3]. The routing solution can be built using ant-based agents behavior in their foraging of network states. These collective agents indirectly communicate through pheromone trailing (stigmergy) in the environment. By following the pheromone trail of another, an agent can find a "good" route in terms of shortest, least congested path from the source to the destination to route the network data. Two basic algorithms are ant-based control (ABC) for telephone networks, which was proposed by Schoonderwoerd et al. [4] and AntNet for packet switching networks, which was proposed by Di Caro et al. [5]. Some subsequent enhancement schemes to improve the ant-based routing performance include smart agents which use dynamic programming [9], reinforcement learning which enhances the ant's adaptability to its environment [10], and a genetic algorithm which adapts the ant control parameters to the search process [11]. While the

above research focuses on the routing problem in electronic communication networks, our interest in this paper is the dynamic RWA problem in WDM optical networks with the constraint of wavelength continuity.

Valera et al. [12] proposed an ACO approach for solving the static RWA problem. The goal is to minimize the number of wavelength requirement given a network topology and a traffic matrix. The wavelength assignment simply uses a greedy method that assigns the lowest available wavelength to each link. An ant's route is selected based on the weight of attraction of each link. Each ant has its own pheromone that can be repulsed by others. Each ant keeps a "tabu" list of previously visited node for route backtracking and loop avoidance. The pheromone updating can use different methods; the best result of this approach is obtained through global update when the weight of attraction of ant for a path increases with the number of traversed ants. The result can be compared to the conventional Nagatsu heuristic [13], but it requires a much longer computational time. However, this approach cannot be applied directly to the dynamic RWA problem.

Garlick et al. [14] proposed an ACO-based algorithm to solve the dynamic RWA problem. When a new connection request arrives, a number of ants are launched from the source to the destination. Ants evaluate a path based on its length and the mean available wavelengths along the path. Global pheromone updating is performed when an ant reaches its destination. The pheromone updating is on a per-demand basis: the network pheromone matrix is reset once a connection is established. The final best path for a connection request is made when all ants complete their exploitation tasks. The authors showed that this algorithm has better performance than an exhaustive search over all available wavelengths for the shortest path [15]. As a new set of ants must be launched for each new connection request, the setup delay will be very high due to the waiting for ants in large networks. In fact, this approach does not show the collective behavior of ants that come from different requests, which is an important aspect of ant-based systems.

3 Ant-Based RWA Algorithm

An optical WDM network is represented by a graph with N nodes and E links. We assume that each link is bi-directional with a capacity of W wavelengths and no nodes have a wavelength conversion capability (wavelength continuity constraint). In order to support the route selection by ants, each network node has a routing table with $N-1$ entry. In a node i with k_i neighbors, the routing table has a k_i column. Each entry corresponds to a destination node and each column corresponds to a neighbor node. The value $r_{n,d}^i$ is used as the selection probability of neighbor node n when an ant is moving towards its destination node d . In order to support the wavelength assignment, we introduce the selection probability of each wavelength into the routing table. For each neighbor node, let P_j be the probability that an ant selects the wavelength j when it moves to destination d .

An example of the new routing table when $W=2$ is shown in Fig.1. When a connection request occurs between source node 1 and destination node 6, node 3 will be selected as next hop because $r_{2,6}^1 < r_{3,6}^1$. Wavelength 2 is preferred over wavelength 1 because $P_1 < P_2$ in that case.

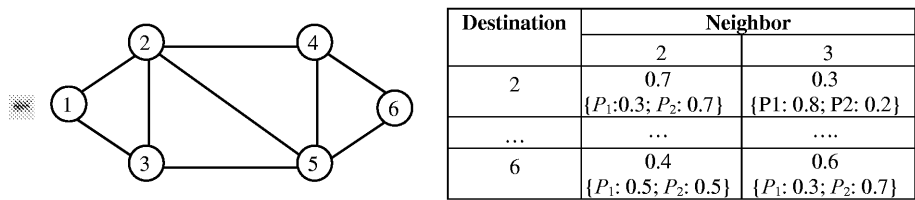


Fig. 1. A network and its routing table from node 1

On a node, ants are launched with a given probability ρ to a randomly selected destination every T time units. Here ρ and T are design parameters. Each ant is considered to be a mobile agent: it collects information on its trip, performs routing table updating on visited nodes, and continues to move forward as illustrated in Fig.2.

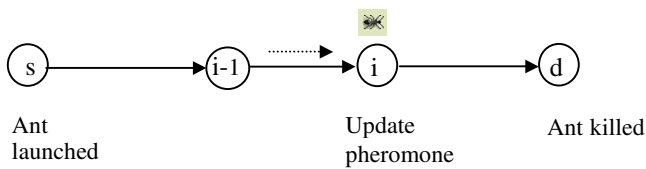


Fig. 2. Ant’s moving and updating tasks

An ant moves from a source to a destination, node by node on a selected wavelength. Its next hop is determined stochastically: a neighbor is selected based on its selecting probabilities in the routing table. An ant is killed when it reaches its destination node or when it cannot select a free wavelength on the selected path for its next move. To avoid a “frozen” status in which all ants concentrate on one route (stagnation), a random scheme is introduced: each ant selects its next hop randomly with an exploiting probability (P_{noise}). When a connection request arrives, the path will be determined based on the highest selection probability node among neighbor’s entries. The wavelength assignment is based on the wavelength selection probabilities from the routing table, or some others conventional heuristics can be used.

Whenever an ant visits a node, it updates the routing table element with the information gathered during its trip. The principle of pheromone update is described as follows. Suppose an ant moves from source s to destination d following the path $(s, \dots, i-1, i, \dots, d)$. When the ant arrives at node i , it will update the entry corresponding to the node s : the probability of neighbor $i-1$ is increased while the probabilities of others neighbors is decreased. For the last visited neighbor $i-1$, the probabilities corresponding to free wavelengths are increased, whereas the probabilities corresponding to busy wavelengths are decreased. More formally, suppose that at time t , an ant visits node i , so the values for routing entry in next time $t+1$ are determined by the following formula (remember that the sum of probabilities for all neighbors is always 1):

$$r_{i-1,s}^i(t+1) = \frac{r_{i-1,s}^i(t) + \delta r}{1 + \delta r} \tag{1}$$

$$r_{n,s}^i(t+1) = \frac{r_{n,s}^i(t)}{1 + \delta r}, n \neq i-1 \quad (2)$$

As described in a previous work [9], smart agents can efficiently improve the performance of ant-based routing systems. Based on the idea of smart agents, the pheromone updating will affect not only the entry corresponding to the source node, but also will affect all the entries corresponding to previous nodes along the path. In order to facility smart updating, an ant must push the information about visited nodes into its stack: node identification, a binary mask that determines the status of free wavelengths on the links it traversed (this mask has W bits corresponding to the number of wavelengths). This stack also serves for loop detection and backtracking, to ensure that ants will not move forever on the network.

The reason for using a wavelength mask is that under wavelength continuity constraint, the number of free wavelengths (congested information) can be found exactly along a path; this will enhance the performance of the ACO approach. At each node, the wavelength mask is updated as below:

$$M_{ant} = M_{ant} \otimes M_{link} \quad (3)$$

where M_{ant} is the actual mask for available wavelengths and M_{link} is the mask for available wavelengths on the next selected link. The amount of trailing pheromone left on a node by an ant depends on the path length and the number of free wavelengths along the path; this amount is computed as below:

$$\delta r = \frac{\alpha}{\delta l} + (1 - \alpha) * \delta w \quad (4)$$

where δl is the increased amount of pheromone corresponding to the length of the path. δw is the increased amount corresponding to the percent of free wavelengths of the path that ant moved along, and α is a scalar parameter that can be used to adjust the emphasis of path length versus free wavelength percentage. For the normalization of pheromone updating, these two factors are computed by the following formula:

$$\delta l = \beta * (e^{\frac{-1.0}{l}} - e^{-1}), \delta w = \gamma * (e^w - e^0) \quad (5)$$

where w is the percentage of free wavelengths of the path and w is deducted from the corresponding wavelength mask in ant stack. Eq.5 guarantees that ants will tend to follow the shorter and less congested path. Here α , β and γ are design parameters which can be adjusted to get the best system performance.

Pheromone updating for wavelength selection probability is based on a similar principle. But instead of updating for the selected wavelength that the ant used, all the free wavelengths along the path are updated. Suppose that k is the index of wavelength for entry $r_{n,d}^i$ in the routing table, we have:

$$P_k = \frac{P_k + \frac{1}{W}}{1 + \frac{|M_{ant}|}{W}} \quad \text{if} \quad M_{ant}(k) = 0 \quad (free) \quad (6)$$

$$P_k = \frac{P_k}{1 + \frac{|M_{ant}|}{W}} \quad \text{if} \quad M_{ant}(k) = 1 \quad (busy) \quad (7)$$

Eqs.6 and 7 guarantee that $\sum_{k=1}^W P_k = 1$ (the normalization condition for wavelength selection probability), and they also ensure that the amount of increased pheromone is proportional to the number of free wavelengths. For the wavelength assignment, we use a simple heuristic: the wavelength with the highest probability among the free wavelengths will be selected.

Our algorithm is briefly described as follows:

{Ant generation}

Do

For each node in network

Select a random destination;

Launch ants to this destination with a probability

End for

Increase time by a time-step for ants' generation

Until (end of simulation)

{Ant foraging}

For each ant from source s to destination d do (in parallel)

While current node $i \neq d$

Update routing table elements

Push trip's state into stack

If (found a next hop)

Move to next hop

Else

Kill ant

End if

End while

End for

{Routing and Wavelength Selection}

For each connection request do (in parallel)

Select a path with highest probability

Search a free wavelength with highest probability

If (found)

Setup a lightpath

Else

Consider a blocking case

End if

End for

4 Simulation Results and Analysis

An extensive experimental study based on Network Simulator *ns-2* [16] has been performed to validate our new ant-based algorithm for RWA. As the original *ns-2* supports packet switching, this feature was used to simulate the ants' moves. We suppose that the control plane for optical WDM networks is implemented in an electronic net-

work that has a same topology as the optical network. An optical routing module was added into *ns-2* to simulate our RWA algorithm.

We used the fixed routing scheme with shortest path algorithm for performance comparison. All the tests were conducted based on the *NSF* network topology with 14 nodes and 21 links as shown in Fig.3, and $W=8$ and $W=16$ were considered in our experiments.

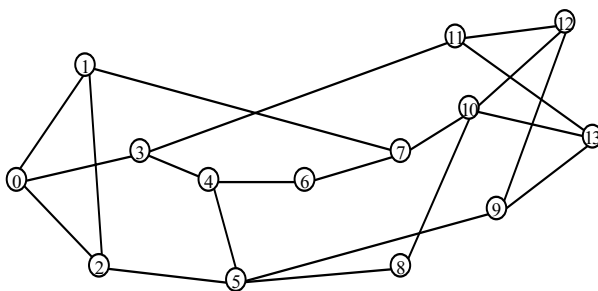


Fig. 3. The NSF Network Topology

We use a dynamic traffic model in which connection requests arrive at each node according to a Poisson process with an arrival rate λ (call/s). The arriving sessions are randomly distributed over the network. The session holding time is exponentially distributed with mean μ (The mean session holding time is μ seconds). If there are N sessions all over the network, then the total network load is measured by $N*\lambda*\mu$ (Erlands). In our experiments, a total $N=50$ traffic session requests are randomly distributed over the networks, thus λ and μ is modified to have different values of workload.

The control parameters for the pheromone updating α , β , γ are selected in advance in each experiment. To control the number of generated ants on the network, an ant is launched from each node to a random destination every T seconds with a probability ρ , where T is pre-selected. The number of ants must be large enough to collect and update the network state into routing tables. If ρ is too small, the routing tables are not changed according to congested state, thus increasing the blocking probability. If ρ is too large, the ants will degrade overall system performance. Thus ρ must be tuned to obtain a best system performance in each experiment. In order to find an initial routing solution, the ants are launched in the network at an initial time. During this period, no traffic arrives on the network. In this case, the value $\alpha=1.0$ is used. As the result, only the path length is taken into account when ants update the routing tables. Thus, the shortest path between every pairs of node is found after the initial period of our algorithm (A formal demonstration of shortest path solution using an ant colony is described in [3]).

Fig.4 shows the relation between ρ and the blocking probability when the network load is 60, 70 and 80 Erlands with $W=8$ wavelengths (Here $T=10ms$, $\alpha=0.3$, $\beta=50$, $\gamma=0.2$, $P_{noise}=0.06$). It is found that when ρ is small, the number of ants is not large enough to collect the network changing states, thus the blocking probability is high. When ρ increases, a larger amount of ants are launched to exploit and update the routing table, thus the blocking probability is reduced. However, when the value of ρ is

very large, the blocking probability cannot be reduced, in fact, it can be increased again due to the performance degradation caused by a very large number of ants on the network. In that case, the new algorithm cannot perform well because the system's performance has been degraded by the ant's stimulation.

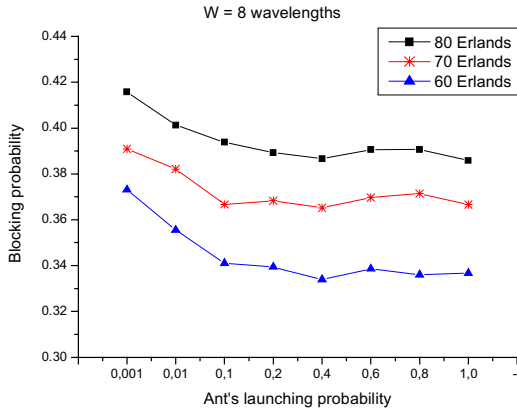


Fig. 4. Blocking probability vs. ρ with $T = 10\text{ms}$

The blocking probabilities on five traffic levels (60-100 Erlands) are presented in Fig.5.a. (with $W=8$) and Fig.5.b. (with $W=16$). The parameters of our algorithm are: $T=10\text{ms}$, $\alpha=0.3$, $\beta=50$, $\gamma=0.2$, $P_{\text{noise}}=0.06$. For every 10ms, an ant is launched from a source with a tunable probability ρ for each traffic level. For each case, the ant's launching probability ρ is adjusted to get a lowest blocking probability. Fixed routing using shortest path and First Fit wavelength assignment schemes are used for comparison.

Fig. 5 shows that the new ant-based algorithm is better than the fixed routing algorithm in terms of blocking probability. The performance is slightly different when the number of wavelength is small ($W=8$) and it is significantly different when we increase this number to $W=16$. These results demonstrate clearly that our ant-based algorithm can always improve routing performance in terms of blocking probability, and this improvement becomes more significant for networks with a larger number of wavelengths per link.

5 Conclusion and Future Works

In this paper, we have proposed an ant-based mobile agents approach to solving the routing and wavelength assignment problems in dynamic WDM networks. We developed a new routing table structure and also a way to adapt the routing table according to network state, using a suitable number of ants that continuously exploit the network. Our simulation shows that the new ant-based algorithm outperforms the fixed-routing algorithm using shortest path and First Fit wavelength assignment scheme. An advantage of this new algorithm is that the path for a connection request is determined

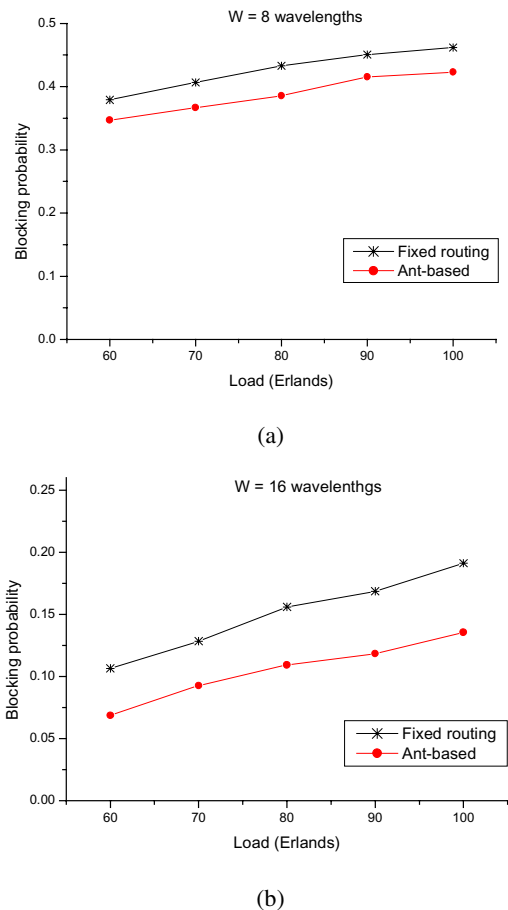


Fig. 5. Comparisons between new Ant-based algorithm and Fixed routing algorithm. (a) Comparison results when $W = 8$. (b) Comparison results when $W = 16$.

immediately on arrival, based on the adapting routing table, so the setup delay time is significantly reduced compared to the fixed routing scheme. Our new algorithm is very flexible in the sense that the number of ants in the network can be efficiently controlled by simply adjust the launching probability of ants to achieve the best performance.

In our future work, we will extend this algorithm by using a reinforcement learning approach such that others ACO control parameters could be automatically adjusted for a given network condition. The other heuristics for routing and wavelength assignment with wavelength conversion will also be investigated.

Acknowledgement. This research is partly supported by the Grand-In-Aid of scientific research (B) 14380138 and 16700056, Japan Science Promotion Society.

References

1. Ramaswami, R., Sivarajan, K.N.: Routing and wavelength assignment in all-optical networks. *IEEE/ACM Transactions on Networking*, vol. 3 (1995) 489-500.
2. Zang H. et al.: A review of routing and wavelength assignment approaches for wavelength-routed optical WDM networks. *Optical Networks Magazine*, vol. 1, no. 1 (2000) 47-60.
3. Bonabeau, E. et al.: *Swarm intelligence: from natural to artificial systems*. Oxford University Press, Inc., New York (1999)
4. Schoonderwoerd, R. et al.: Ant-like agents for load balancing in telecommunications networks. *Proc. of the First International Conference on Autonomous Agents*. ACM Press, (1997) 209-216.
5. Di Caro, G., Dorigo, M.: *AntNet: A mobile agents approach to adaptive routing*. Technical Report 97-12, IRIDIA, Universite Libre de Bruxelles (1997)
6. Ramamurthy, S., Mukherjee, B.: Fixed-Alternate Routing and Wavelength Conversion in Wavelength Routed Optical Networks. *Proc. IEEE Globecom* (1998)
7. Li, L., Somani, A.K.: Dynamic wavelength routing using congestion and neighborhood information. *IEEE/ACM Trans. on Networking*, vol. 7, no. 5 (1999) 779-786.
8. Kassabalidis, I. et al.: Swarm intelligence for routing in communication networks. *Proc. IEEE Globecom* (2001)
9. Bonabeau, E. et al.: Routing in Telecommunication Networks with Smart Ant-Like Agents. *Proc. IATA, Lectures Notes in AI*, vol. 1437, Springer Verlag (1998)
10. Legge, D. Baxendale, P.: An Agent-Managed Ant-Based Network Control System. *AAMAS-3* (2003)
11. White, T. et al.: ASGA: Improving the Ant System by Integration with Genetic Algorithms. *Proc. GP/SGA* (1998)
12. Navarro Varela, G., Sinclair, M.C.: Ant Colony Optimization for Virtual-Wavelength-Path Routing and Wavelength Allocation. *Proc. CEC'99*, Washington DC, USA (1999).
13. Nagatsu, N. et al.: Number of wavelengths required for constructing large-scale optical path networks. *Electronics and Comm. in Japan*, part 1, vol. 78, no. 9 (1995) 1-11.
14. Garlick, R.M., Barr, R.: Dynamic wavelength routing in WDM networks via Ant Colony Optimization. *Ant Algorithms*, Springer-Verlag Publishing (2002) 250-255
15. Mokhtar, A., Azizoglu, M.: Adaptive wavelength routing in all-optical networks. *IEEE Trans. on Networking*, vol. 6 (1998) 197-206.
16. The Network Simulator, ns-2. <http://www.isi.edu/nsnam/ns/index.html> (2003)